Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/27/2005
PI Name:	Maris, Humphrey J. Ph.D.		
Project Title:	Coalescence of Superfluid Helium Drops in a Microgravity Environment		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	401-863-2185
Organization Name:	Brown University		
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City:	Providence	**State:**	RI
Zip Code:	02912	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	05/01/2001	**End Date:**	03/31/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity		
Task Description:	We will conduct ground-based research to study the coalescence of levitated drops of liquid helium. The drops will be levitated using the superconducting magnet levitation system developed in our laboratory. The coalescence of liquid drops will be studied as a function of impact parameter, impact velocity, temperature, and drop size. The effect of these parameters on the surface morphology during the coalescence process will be investigated. A high speed line camera will be used to carry out an investigation of the initial rate of growth of the neck between the two drops. The results of this research will provide necessary data for the design of an experiment to study coalescence of large superfluid helium drops in space. The long-term objective of this research is the understanding of the process of coalescence of liquid drops. The unique properties of helium make it an ideal material for coalescence studies. The hydrodynamics of helium in the superfluid phase is described by the two-fluid model of Landau and Tisza. At sufficiently low temperatures where		

	the density of the normal fluid becomes extremely small, liquid helium acts as an ideal fluid with no viscosity. Our long term goal is a space experiment to study the coalescence of superfluid helium in this low temperature regime. In a space experiment large helium drops could be used, and this would make it possible to resolve the early stages of the coalescence process to a degree that is impossible in a ground based experiment. The study of coalescence is of great importance in a number of practical applications, including the sintering process, the combustion of fuel sprays, and the process of spray painting. In nature, coalescence affects the distribution of size of rain drops.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	During the past year we have completed a study of the coalescence of two classical drops. The drops were high-viscosity silicon oil suspended in a water-alcohol mixture of nearly equal desnity. We made careful measurements of the evolution of the shape of the drops after coalescence, and performed a detailed study of the variation of the radius of the neck between the drops as a function of time. We compared the results with theoretical predictions. We have completed a writeup of this work and this paper has just appeared in Physical Review E (see Bibliography section).
Bibliography Type:	Description: (Last Updated: 01/07/2020)
Articles in Peer-reviewed Journals	Yao W, Pennington P, Maris HJ, Seidel GM. "Coalescence of viscous liquid drops." Phys Rev E Stat Nonlin Soft Matter Phys. 2005 Jan;71(1 Pt 2):016309. https://doi.org/10.1103/PhysRevE.71.016309 ; PMID: 15697724 , Jan-2005